

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EC)

May 2013

EC 407 Radar Systems

Date: 11.05.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain basic principle of radar. [3]
(b) Derive basic Radar equation. [4]
- Q - 2 (a) What types of display use in radar? Explain any two in detail. [6]
(b) Explain types of duplexer in detail. [5]
(c) Explain how multiple staggered PRF's can be processed with a transversal filter. [3]

OR

- Q - 2 (a) What is mixer? Explain Balanced mixer. [5]
(b) What are important applications of radar? Also describe frequency band used for various application. [5]
(c) An MTI radar is operating at 8 GHz with a PRF of 1200 Hz. Calculate the lowest three blind Speeds. [4]
- Q - 3 (a) Write a short note on Distance measuring equipment. [6]
(b) Explain Coherent MTI radar with the help of block diagram. What is the need of COHO and STALO? [5]
(c) Explain DECCA. [3]

OR

- Q - 3 (a) Explain basic principle of MTI radar. [5]
(b) Two MTI radars are found to have blind speeds are given by $V_{b2}=25\text{m/s}$ of first MTI radar and $V_{b3}=35\text{m/s}$ of second MTI radar. If they operate at a wavelength of 3 cm, then find the ratio of PRFs of the Radars. [5]
(c) Explain Airborne radar. [4]

SECTION - II

- Q - 4 (a) Explain the following terms: [3]
(i) navigation (ii) Radar Aiding (iii) Corse Aiding
- (b) Draw neat block diagram of FMCW radar and enlist its distinguished features. [4]
- Q - 5 (a) How Detection of multiple targets moving with different velocities had done by FMCW radar? [5]
(b) Enlist various errors in direction finding. Explain one of them in detail. [5]
(c) Explain the synthetic aperture radar. [4]

OR

- Q - 5 (a) List out the various methods used for increased loop voltage. Explain any two in brief. [6]
(b) What are the limitations of direction finder? Explain the operation of sense finder? [5]
(c) Draw the block diagram of Automatic direction finder. [3]
- Q - 6 (a) Explain the operation of four course radio range radar. [5]
(b) Write a short note on Instrument Landing System Radar. [5]
(c) Write a short note on OMEGA. [4]

OR

- Q - 6 (a) Describe the principle of VOR. Derive the equation of electric field of a VOR System with five antenna configuration. [5]
(b) Explain the salient features of TACAN. [5]
(c) Write a short note on Microwave landing system. [4]